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Candidate Number

HONG KONG DIPLOMA OF SECONDARY EDUCATION EXAMINATION BYTOPIC

MATHEMATICS Extended Part

Module 2 (Algebra and Calculus)

Question-Answer Book

Time allowed : $2\frac{1}{2}$ hours

This paper must be answered in English

INSTRUCTIONS

(1) After the announcement of the start of the examination, you should first write your Candidate Number in the space provided on Page 1 and stick barcode labels in the spaces provided on Pages 1.

(2) This paper consists of TWO sections, A and B.

(3) This paper carried 100 marks.

(4) Attempt ALL questions in this paper. Write your answers in the spaces provided in this Question-Answer Book. Do not write in the margins. Answers written in the margins will not be marked.

(5) Graph paper and supplementary answer sheets will be supplied on request. Write your Candidate Number, mark the question number box and stick a barcode label on each sheet, and fasten them with string INSIDE this book.

(6) Unless otherwise specified, all working must be clearly shown.

(7) Unless otherwise specified, numerical answers must be exact.

(8) The diagrams in this paper are not necessarily drawn to scale.

(9) No extra time will be given to candidates for sticking on the barcode labels or filling in the question number boxes after the 'Time is up' announcement.

FORMULAS FOR REFERENCE

$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$	$\sin A + \sin B = 2\sin\frac{A+B}{2}\cos\frac{A-B}{2}$
$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$	$\sin A - \sin B = 2\cos\frac{A+B}{2}\sin\frac{A-B}{2}$
$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$	$\cos A + \cos B = 2\cos\frac{A+B}{2}\cos\frac{A-B}{2}$
$2\sin A \cos B = \sin(A+B) + \sin(A-B)$	$\cos A - \cos B = -2\sin\frac{A+B}{2}\sin\frac{A-B}{2}$
$2\cos A \cos B = \cos(A+B) + \cos(A-B)$	
$2\sin A \sin B = \cos(A-B) - \cos(A+B)$	

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SECTION A (56 marks)

1. (a) Prove that $1 + \dfrac{2\sin 2x - \cos 3x}{\cos x} = 4\sin x(1 + \sin x)$ for all x where $\cos x \neq 0$. (3 marks)

   (b) Solve $2 + \dfrac{2\sin 2\theta - \cos 3\theta}{\cos\theta} = 0$, where $-\pi \leq \theta \leq \pi$. (3 marks)

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2. (a) Prove that $\cos A + \cos B + \cos C + \cos(A+B+C) = 4\cos\frac{A+B}{2}\cos\frac{B+C}{2}\cos\frac{C+A}{2}$. (3 marks)

(b) Solve $\cos x + \cos 3x + \cos 5x + \cos 9x = 0$, where $-\frac{\pi}{3} < x < \frac{\pi}{6}$. (3 marks)

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3. (a) Prove that $\tan x + 2\tan 2x = \cot x - 4\cot 4x$. (3 marks)

(b) Solve $\tan x + 2\tan 2x + 4\tan 4x + 8\tan 8x - \cot x = 16$, where $\frac{\pi}{10} < x < \frac{\pi}{8}$. (3 marks)

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4. It is given that $0 \le \theta \le \frac{\pi}{2}$ and $\sin 2\theta = \frac{2}{3}$.

(a) Without finding the value of θ, prove that

$$\tan^2\theta + \cot^2\theta = 7.$$

(3 marks)

(b) Find the exact value of

$$\frac{1}{1+\tan^2\theta} + \frac{1}{1+\cot^2\theta} - \frac{12}{\tan^4\theta + \cot^4\theta - 2}.$$

(3 marks)

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5. (a) Expand $(a+b+c)^2$. (1 mark)

(b) Let α, β, γ be real numbers such that $\cos\alpha+\cos\beta+\cos\gamma=0$ and $\sin\alpha+\sin\beta+\sin\gamma=0$.

(i) Find the value of $\cos(\alpha-\beta)+\cos(\beta-\gamma)+\cos(\gamma-\alpha)$. (3 marks)

(ii) Prove that $\cos 2\alpha+\cos 2\beta+\cos 2\gamma=0$. (2 marks)

(c) Let ϕ be a real constant. Find the value of

$$\sin(\alpha-\phi)\cos(\alpha-\phi)+\sin(\beta-\phi)\cos(\beta-\phi)+\sin(\gamma-\phi)\cos(\gamma-\phi).$$

(3 marks)

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6. It is given that θ is an acute angle satisfying the equation

$$\tan^6\theta - 33\tan^4\theta + 27\tan^2\theta - 3 = 0.$$

(a) (i) Express $\tan 3x$ in terms of $\tan x$. (2 marks)

(ii) Prove that $\tan^2 3\theta = 3$. (3 marks)

(b) Hence, find all possible values of θ in radians. (2 marks)

(c) Using the result of (b), find the product of all possible values of θ without using a calculator. (3 marks)

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7. Let n be a positive integer. It is given that for all real numbers θ satisfying $\sin\theta \neq 0$,

$$\frac{\sin(2n+1)\theta}{\sin\theta} = 1 + 2\sum_{k=1}^{n}\cos(2k\theta).$$

(a) Prove that $\cos\frac{2\pi}{7} + \cos\frac{4\pi}{7} + \cos\frac{6\pi}{7} = -\frac{1}{2}$. (2 marks)

(b) (i) Prove that $\cos 3x = 4\cos^3 x - 3\cos x$. (2 marks)

(ii) By considering the equation $\cos 3\theta = \cos 4\theta$, prove that $8\cos^3\theta + 4\cos^2\theta - 4\cos\theta - 1 = 0$ holds for $\theta = \frac{2\pi}{7}$. (3 marks)

(iii) Hence, without using a calculator, find the value of $\cos\frac{2\pi}{7}\cos\frac{4\pi}{7}\cos\frac{6\pi}{7}$. (2 marks)

(c) Using the results of (a) and (b), find the value of each of the following without using a calculator:

(i) $\cos^2\frac{2\pi}{7} + \cos^2\frac{4\pi}{7} + \cos^2\frac{6\pi}{7}$ (2 marks)

(ii) $\sec\frac{2\pi}{7} + \sec\frac{4\pi}{7} + \sec\frac{6\pi}{7}$ (2 marks)

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SECTION B (44 marks)

8. (a) By mathematical induction, prove that for all positive integers n,

$$\sum_{k=1}^{n} \frac{1}{2^k} \tan\left(\frac{\theta}{2^k}\right) = \frac{1}{2^n} \cot\left(\frac{\theta}{2^n}\right) - \cot\theta,$$

where $\theta \neq m\pi$ (m is an integer). (4 marks)

(b) Hence, find the exact value of

$$\frac{1}{32} \tan\left(\frac{\pi}{64}\right) + \cdots + \frac{1}{1024} \tan\left(\frac{\pi}{3072}\right).$$

(2 marks)

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9. (a) Prove that $\tan\left(\frac{\pi}{4}-\theta\right)=\frac{\cos 2\theta}{1+\sin 2\theta}$, where $\sin 2\theta \neq -1$ and $\cos\theta \neq 0$. (2 marks)

(b) Let ϕ be a real number. Prove that for all positive integers n,

$$\sum_{r=1}^{n} \mathrm{C}_r^{n+1}(-1)^r(\sin 2\phi)^r(1+\sin 2\phi)^{n+1-r}=1-(1+\sin 2\phi)^{n+1}-(-\sin 2\phi)^{n+1}.$$

(2 marks)

(c) Let $\phi=\frac{\pi}{8}$. Using the results of (a) and (b), find the exact value of the series

$$\sum_{r=1}^{3} \mathrm{C}_r^{4}(-\frac{1}{2})^r\,(\sin 4\phi)^r\,(\cos 2\phi)^{4-2r}\left[\tan\left(\frac{\pi}{4}-\phi\right)\right]^{r-4}.$$

(3 marks)

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10. It is given that θ is an acute angle. Let $I = \int \frac{1}{\sin\theta(2+\cos\theta)}\,d\theta$.

(a) Prove that

$$\frac{1}{\sin\theta(2+\cos\theta)} = \frac{\sin\theta}{6(1-\cos\theta)} + \frac{\sin\theta}{2(1+\cos\theta)} - \frac{\sin\theta}{3(2+\cos\theta)}.$$

(2 marks)

(b) Prove that

$$I = \frac{1}{3}\ln\left(3\tan\frac{\theta}{2} + \tan^3\frac{\theta}{2}\right) + C,$$

where C is an arbitrary constant. (4 marks)

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11. Let $f(x)=\sqrt{\frac{1-\sin 2x}{1+\sin 2x}}$, where $0<x<\frac{\pi}{4}$.

(a) Prove that $f(x)=\frac{\cos x-\sin x}{\cos x+\sin x}$. (2 marks)

(b) From first principles, find $f'(x)$. (4 marks)

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12. Let O be the origin. In a 3-dimensional space, $\vec{a}$ and $\vec{b}$ are two non-parallel unit vectors such that the angle between them is $\frac{\pi}{3}$. Two variable vectors $\vec{u}$ and $\vec{v}$ are defined for a real number θ ($0 < \theta < \frac{\pi}{2}$) by:

$$\vec{u} = (\cos\theta)\vec{a} + (\sin\theta)\vec{b} \quad \text{and} \quad \vec{v} = (\sin\theta)\vec{a} - (\cos\theta)\vec{b}.$$

(a) Prove that $\vec{u} \times \vec{v} = -(\vec{a} \times \vec{b})$. (2 marks)

(b) Let Π be the plane spanned by the vectors $\vec{u}$ and $\vec{v}$. It is given that P is a fixed point in space with position vector $\vec{OP} = \vec{a} + 2\vec{b} + (\vec{a} \times \vec{b})$.

(i) Find the perpendicular distance d from point P to the plane Π. (3 marks)

(ii) Prove that d is a constant independent of θ. (1 mark)

(c) Another variable vector is given by $\vec{w} = \vec{u} + \tan\theta\,\vec{v}$.

(i) Express $|\vec{w}|^2$ in terms of trigonometric functions of θ. (3 marks)

(ii) Using differentiation, find the minimum value of $|\vec{w}|^2$ in the interval $0 < \theta < \frac{\pi}{2}$. (2 marks)

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13. Let $M = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$ and $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, where $0 < \theta < \frac{\pi}{2}$.
It is given that $\det(M^2 - \sqrt{3}M + I) = 0$.

(a) Prove that $2\cos 2\theta - 4\sqrt{3}\cos\theta + 5 = 0$. (3 marks)

(b) Hence, find the exact value of θ. (3 marks)

(c) Let M be the matrix with the value of θ found in (b). A sequence of matrices is defined by $M^n = a_n M + b_n I$ for all positive integers n, where a_n and b_n are real constants. Find the exact value of a_{2026}. (2 marks)

END OF PAPER

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